

[DASHBOARD](#) / [CORSI](#) / [LANGUAGES AND ALGORITHMS FOR ARTIFICIAL INTELLIGENCE](#) / [SEZIONI](#) / [MODULE 3](#) / [PROBLEM EXAMPLES](#)

Iniziato	mercoledì, 16 giugno 2021, 12:10
Stato	Completato
Terminato	mercoledì, 16 giugno 2021, 17:43
Tempo impiegato	5 ore 32 min.
Valutazione	Non ancora valutato

Domanda **1**

Completo

Punteggio
max.: 1,00

Construct a deterministic TM of the kind you prefer, which decides the following language:

$$\mathcal{L} = \{w \in \{0,1\}^* \mid \text{w contains } 10 \text{ as a substring}\}.$$

Study the complexity of TM you have defined.

Single-tape TM:

$(q_{\text{init}}, \triangleright) \rightarrow (q_0, \triangleright, R)$
 $(q_0, 0) \rightarrow (q_1, 0, R)$
 $(q_0, 1) \rightarrow (q_2, 1, R)$
 $(q_1, 0) \rightarrow (q_1, 0, R)$
 $(q_1, 1) \rightarrow (q_2, 1, R)$
 $(q_2, 0) \rightarrow (q_{\text{halt}}, 0, S)$
 $(q_2, 1) \rightarrow (q_2, 1, R)$

The TM above take linear time in the input length, as it scans—at most—the whole string one symbol at a time.

Domanda **2**

Completo

Punteggio
max.: 1,00

You are required to prove that the following function f is in **FP**. To do that, you can give a TMs or define some pseudocode. The function is one that, given two strings $v, w \in \Sigma^*$ checks whether v is a substring of w , namely whether w contains an exact occurrence of v .

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INPUT:  $v, w$  in  $\Sigma^*$ 
OUTPUT: 1 if  $v$  is a substring of  $w$ , 0 otherwise
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for  $i \leftarrow 1$  to  $|w| - |v| + 1$  do
  if  $w[i : |v| + i - 1] = v$  then
    return 1
return 0

```

1. INPUT ENCODING

We consider as input length $n = |v| + |w|$. Since w contains $O(|w|) = O(n)$ different symbols (from Σ), encoding those symbols requires $O(\lg|w|) = O(\lg n)$ bits, hence $|_L w| = O(n \lg n)$; the same holds for v . Therefore, $l = |_L(v, w)| = O(n \lg n)$, which is polynomial in n .

2. NUMBER OF INSTRUCTIONS

The number of instructions is dominated by the for loop, which is executed $O(|w|) = O(n) = O(n \lg n) = O(l)$, so it is polynomial in l .

3. EXECUTION TIME OF EACH INSTRUCTION

The only nontrivial instruction is a comparison between two strings of length $|v| = O(n) = O(n \lg n) = O(l)$, which can be performed in polynomial time in l .

4. SIZE OF INTERMEDIATE RESULTS

i is a natural number $< |w|$, so $|_L i| = O(\lg|w|) = O(\lg n) = O(n \lg n) = O(l)$, which is polynomial in l .

Domanda **3**

Completo

Punteggio
max.: 1,00

We studied the classes SAT, and 3SAT, and 2SAT. You are required to classify the problem 1SAT containing all (and only) the satisfiable 1CNFs. In particular, to which complexity class does 1SAT belong? Prove your claim.

A 1CNF is a CNF such that each clause has ≤ 1 literals, i.e. it is a conjunction of literals. Therefore, a 1CNF is satisfiable iff all its literals can be true, i.e. it does not contain two literals that are one the negation of the other.

INPUT: $L = [l_1, \dots, l_n]$, the list of literals in a 1CNF
OUTPUT: 1 if the 1CNF is satisfiable, 0 otherwise

```

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BOOL_VARS ← [ ]
for i ← 1 to n do
  if  $\neg l[i]$  in BOOL_VARS then
    return 0
  BOOL_VARS.append( $l[i]$ )
return 1

```

1. INPUT ENCODING

We consider n as the input length, i.e. the number of literals in the 1CNF. A literal, by definition, is either a propositional symbol or its negation. Therefore in L there are at most as many propositional symbols as there are literals, i.e. L contains $O(n)$ propositional symbols. A first intermediate encoding of L over $\{0, 1, \neg, \#\}^*$ thus requires $O(n)$ binary encodings of $O(n)$ different symbols, which take $O(\lg n)$ bits each, $O(n)$ $\neg$ s, and $O(n)$ $\#$ s. The final binary encoding has therefore length $|L| = O(n \lg n)$.

2. NUMBER OF INSTRUCTIONS

The number of instructions is dominated by the for loop, which is executed $O(n) = O(n \lg n) = O(|L|)$ times, so polynomial in $|L|$.

3. EXECUTION TIME OF EACH INSTRUCTION

The only nontrivial instruction is the append, which takes time $O(n) = O(n \lg n) = O(|L|)$, which is polynomial in $|L|$.

4. SIZE OF INTERMEDIATE RESULTS

i is an integer number $\leq n$, so $|i| = O(\lg n) = O(n \lg n) = O(|L|)$. BOOL_VARS is at most equal to L itself, so $|\text{BOOL_VARS}| = O(|L|)$. Therefore both i and BOOL_VARS have an encoding length which is polynomial in $|L|$.

◀ [Question Examples](#)

Vai a...